

\chapter{Literature Review}

\section{Smart Grid Fundamentals}

Advanced Metering Infrastructure (AMI), serving as a foundation for Smart Grids (SGs), is a cornerstone of the modern electrical infrastructure, progressively replacing antiquated power systems in both residential and commercial sectors. SGs represent an evolutionary leap in power grids, incorporating bidirectional information and communication technology (ICT) alongside pervasive computing to enhance energy management and distribution. This sophisticated integration fosters enhanced control, heightened efficiency, augmented reliability, and increased safety in energy distribution \cite{yan2012survey}. The United States Department of Energy's modern grid program designates Smart Grids (SGs) as a combination of control methods, integrated communication systems, and advanced sensing technologies within the existing electrical power infrastructure. This innovation allows both consumers and utility providers to better monitor, manage, and predict energy usage patterns.

As depicted in Figure 2.1, the process starts with Bulk Generation, involving large-scale electricity production from various sources \cite{kusiak2017smart}. This electricity is transmitted at high voltage from generation plants to substations \cite{schavemaker2017electrical}, and then distributed at lower voltages to end-users \cite{bayindir2016smart}. Substations convert transmission to distribution voltage \cite{zhang2019security}. The Advanced Metering Infrastructure (AMI) Network facilitates two-way data communication between customers and utilities \cite{depuru2011smart}. Smart Meters (SM) record and report energy usage \cite{zheng2013smart}. Home Area Network (HAN), Building Area Network (BAN), and Industrial Area Network (IAN) manage local energy usage and generation \cite{gungor2013industrial}. Markets enable the buying and selling of electricity \cite{borenstein2018two}, while Grid System Operators ensure the power system's operation \cite{bompard2021classical}. Finally, Service Providers offer diverse services from energy supply to efficiency solutions \cite{strbac2015microgrids}.

SGs facilitate the integration of microgrids and various distributed energy resources (DER), such as solar and wind power, as well as energy storage systems. This combination aims to tackle prevailing energy management challenges, resulting in a robust, self-healing grid system. Moreover, SGs employ demand-side management (DSM) strategies to encourage consumers to consciously adjust their energy consumption. This incentivizes consumers to shift their usage to off-peak hours, thereby reducing consumption rate fluctuations and balancing peak to average energy demand. This shift is essential for achieving objectives such as reducing greenhouse gas emissions, combating global warming, and striving for national energy independence \cite{mcdaniel2009security}. The following section, 2.2.1, delves into the intricacies of Advanced Metering Infrastructure (AMI), which is crucial for collecting information and data from consumers and loads, underpinning the operation of Smart Grids.

\subsection{Overview of AMI}

The integration of Advanced Metering Infrastructure (AMI) represents a pivotal advancement in the evolution of electrical grids. Central to the smart grid (SG) paradigm is the active engagement and empowerment of consumers—a vision articulated by the U.S. Department of Energy in 2008

\cite{USDeptEnergy2008}. AMI is instrumental in providing a dynamic metering ecosystem that equips consumers with critical insights, fostering informed energy usage and enabling proactive participation. Concurrently, utility providers (UPs) benefit from enhanced customer engagement and optimized operational efficiency through sophisticated asset management facilitated by granular metering analytics. AMI serves as a conduit for real-time energy consumption data and grid event notifications to both consumers and UPs, fostering collaborative energy management. This synergy enhances billing accuracy, aids in demand response initiatives, and supports judicious decision-making within SGs. The role of AMI extends beyond mere data relay; it constitutes a critical nexus connecting UPs, consumers, and the broader spectrum of generation and storage assets, underpinned by a suite of integrated technologies encompassing advanced communication systems, intelligent metering devices, Home Area Networks (HAN), Building Area Network(BAN), Industrial Area Network (IAN) and robust interfaces for utility operations and data management. This thesis delves into the appliance consumption patterns in smart homes and the implications of smart meters (one of the components of AMI) within the electrical power distribution landscape.

\subsection{Architectural Elements of AMI}

The AMI comprises a complex network of intelligent devices, including smart meters (SMs), data aggregation units, and Internet of Things (IoT) devices, as well as a robust data management infrastructure, which includes systems such as Meter Data Management Systems (MDMS) and central communication hubs. These components are interconnected through a sophisticated network infrastructure that seamlessly integrates data into various software applications and physical interfaces. AMI's distinct feature is its bidirectional flow of both communication and electricity, enabling a comprehensive overlay of informational and power exchanges, as noted by Rashed Mohassel et al \cite{mohassel2014survey}. The integrated network design and AMI's role within the smart grid are illustrated in Figure 2.2.

\subsection{Smart Metering Devices}

Smart metering devices represent the cutting edge of technological innovation, equipped with sophisticated hardware and software designed for precise data collection and measurement at predetermined intervals. These devices are configured and programmed by system administrators to transmit data to relevant entities at regular intervals. As part of the bidirectional communication framework of the AMI, these Internet of Things (IoT) devices not only receive but also transmit signals to perform designated actions. Moreover, information on utility pricing from UPs enables these devices to adjust energy consumption according to user-defined preferences and operational parameters.

Traditionally, conventional meters were exclusively used for billing consumer energy usage. However, with the emergence of SGs, contemporary SMs have become increasingly prevalent, enhancing the resilience and operational efficiency of power systems that incorporate DER and distributed demand response initiatives, as identified by W. Wang and Lu \cite{wang2013cyber}. SMs are essential components within AMI ecosystems, serving as advanced energy meters that record consumer energy

usage in regular, predetermined intervals. Unlike traditional meters, SMs possess the capability to capture detailed energy usage metrics, such as voltage, current, frequency, and phase angle. These meters collaborate with central data aggregators and communication gateways to ensure power quality and secure data transmission in real-time. Furthermore, SMs possess the ability to both execute and receive control directives, whether locally or remotely. In addition to their role in consumption regulation and system monitoring, SMs also gather diagnostic data on domestic appliances, the distribution grid, and power-related events. They can remotely regulate energy supply to consumers, thus controlling the maximum energy consumption. The data collected by SMs includes consumption figures, timestamps, and unique meter identifiers, as discussed by Depuru, Wang, & Devabhaktuni [depuru2011electricity]. Figure 2.3 illustrates the metering architectures of both conventional power grids and SGs.

The essential features of SMs encompass a wide array of functionalities, such as:

\begin{itemize}

- \item The capacity for real-time data collection and evaluation
- \item The ability to upgrade and operate remotely
- \item Notification of system failures and malfunctions
- \item Implementation of dynamic pricing structures based on time
- \item Monitoring of power quality
- \item Energy trading through net metering
- \item Load scheduling for Demand Side Management (DSM)
- \item Load limiting to encourage demand response
- \item Detection of energy theft and meter tampering using alert systems and sensors
- \item Advancement of energy conservation for the sake of environmental sustainability.

\end{itemize}

\begin{figure}[h!]

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\subsection{Networking Framework for Advanced Metering Infrastructure}

The Advanced Metering Infrastructure (AMI), as depicted in Figure 2.2, is supported by a tripartite networking system consisting of the Home Area Network (HAN), Neighborhood Area Network (NAN), and Wide Area Network (WAN), each possessing unique characteristics and functions:

\begin{itemize}

\item \textbf{HAN:} Within the domestic realm, HAN connects smart meters (SMs), intelligent appliances, local generation units, energy storage systems, and plug-in hybrid electric vehicles (PHEVs), along with their respective control units. To facilitate high bandwidth and low-energy signal transmission, wireless modalities are the most appropriate communication methods for HAN. Protocols such as ZigBee, WiFi operating at 2.4 GHz, HomePlug standards, and IEEE 802.11 are commonly utilized to fulfill these requirements, as indicated by the U.S. Department of Energy \cite{USDeptEnergy2010}.

\item \textbf{NAN:} Comprising data aggregator units and SMs situated within consumer properties, the NAN facilitates the exchange of information within a localized community. The choice of communication technology within the NAN is tailored to suit various operational contexts and may include Power Line Communication (PLC), RS485 serial connections, GPRS/3G mobile networks, or ZigBee interfaces.

\item \textbf{WAN:} Serving as the communication backbone, the WAN provides connectivity between the data aggregation points and the central operations hub. The selection of communication channels within the WAN is predicated upon regional requirements and may encompass PLC, Ethernet, or GPRS/3G networks.

\end{itemize}

Subsequent to data acquisition, the collated information is conveyed to an operations center. This center houses both a Meter Data Management System (MDMS) and a head-end system. The head-end system orchestrates the communication protocols, aggregates, and preserves metering information, and ensures compatibility with internet protocols while interfacing with various devices. In parallel, the MDMS, as the core component of the AMI's management architecture, oversees the monitoring of the distribution network and undertakes the analysis, maintenance, and operational management tasks integral to the functioning of Smart Grids.

\subsection{Operations and Advantages of AMI}

AMI presents numerous benefits and implications that impact consumers, Utility Providers (UPs), and society at large, as acknowledged by the U.S. Department of Energy (2008).

\subsubsection{For Consumers:}

\begin{itemize}

\item Improved Energy Management: AMI offers extensive insights into energy consumption and grid status, empowering consumers to make informed decisions about their energy use. As a result,

consumers can expect enhanced power quality, increased reliability, and precise billing, ultimately leading to reduced utility expenses.

\end{itemize}

\subsubsection{For UPs:}

\begin{itemize}

\item Operational Efficiency and Billing Process Improvement: AMI enables UPs to gain advantages in areas such as operational efficiency and billing processes.

\item Cost Savings: The implementation of Smart Meters (SMs) facilitates automated data transmission and remote firmware updates, eliminating the need for physical meter readings and maintenance. Consequently, operational and labor costs are reduced.

\item Demand Response and Load Management: AMI equips UPs with the ability to monitor grid load continuously, facilitating dynamic pricing and direct load control. This not only reduces peak demand but also mitigates the need for new-generation facilities and eases transmission congestion.

\item Grid Health Monitoring: Real-time monitoring allows UPs to analyze line losses and optimize the electrical power infrastructure.

\item Enhanced Outage Management: Smart metering aids UPs in promptly detecting and locating outages, enabling more efficient repair crew dispatch.

\item Non-Technical Loss Reduction: SMs and data collectors in AMI play a crucial role in identifying and mitigating energy theft, thereby reducing NTLs.

\end{itemize}

\subsubsection{For Society:}

Environmental Sustainability: AMI contributes to a greener environment by enhancing energy delivery and usage efficiency. It promotes the adoption of Distributed Generation (DG) and DER, potentially reducing carbon dioxide emissions significantly, as suggested by Siddiqui \cite{Siddiqui2008GreenGrid}.

\subsection{Security Concerns and Challenges in AMI}

The widespread adoption of SMs across homes and businesses in the United States and initiatives like Tenaga Nasional Berhad (TNB) \cite{TNB2006} pilot project in Malaysia highlight the growing implementation of AMI. However, this expansion brings about significant security challenges. This thesis will delve into these security concerns, emerging from the new power infrastructure, and explore potential mitigation strategies.

\subsubsection{Privacy Concerns for Consumers}

In the SG ecosystem, while consumers collaborate with UPs for efficient energy management, privacy concerns arise due to the necessity of sharing detailed consumption data. Third parties can utilize fine-grained SM data for load profiling, which reveals not only appliance usage but also personal lifestyle patterns. The ability to deduce such detailed information poses privacy risks, ranging from burglary risks to unsolicited marketing and competitive disadvantages for industrial consumers. Instances like the Dutch Parliament's rejection of SG implementation in 2009 due to privacy concerns \cite{cuijpers2013smart} underscore the need for robust privacy protection measures. Establishing a regulatory framework that defines data collection, sharing, and usage policies is crucial to maintaining consumer trust in SG adoption. Moreover, the security and reliability of smart devices must be rigorously assessed by academia, the government, and the energy industry to ensure comprehensive protection against privacy invasions.

\subsection{Energy Theft: A Multi-Billion Dollar Concern}

Energy theft within electrical power distribution systems has long been an international concern. Northeast Group \cite{NortheastGroup2017ElectricityTheft} estimates the global annual cost of energy theft to be approximately \\$96 billion in 2017, with a notable portion occurring in emerging economies like India, Brazil, and Russia. Historically, methods of energy theft involved physically tampering with meters to disrupt accurate measurements. However, with the introduction of Smart Meters (SMs), the nature of energy theft is evolving towards more complex cyber-related attacks, such as remote device hacking and data manipulation. These advanced attacks can lead to minor alterations in energy readings or even large-scale infrastructure threats, as detailed by McDaniel \& McLaughlin \cite{mcdaniel2009security}.

Smart Meters, integral to the Advanced Metering Infrastructure (AMI), are built using common software and hardware, exposing them to typical vulnerabilities found in networked and communication systems. This includes risks such as unauthorized access, distributed denial-of-service (DDoS) attacks, and various forms of malware, making SMs prime targets for cybercriminals \cite{mcdaniel2009security}. The exploitation of these vulnerabilities can lead to significant financial implications, not only in terms of energy theft but also in the costs of replacing compromised meters. Misuse of SMs could also pose serious risks to the power infrastructure, potentially leading to misguided operational decisions by UPs and hiding imminent threats.

The future development of SG is highly dependent on the regulatory frameworks established by governments and UPs. Addressing the security issues introduced by AMI is crucial for transitioning to a more efficient and cost-effective power grid.

\section{Electricity Losses in Distribution Systems}

Energy loss in electrical distribution systems, as defined by Nagi et al. \cite{nagi2009nontechnical}, refers to the discrepancy between the energy supplied and that reported by consumers. Assessing these losses is key to evaluating system performance. Generally, UPs face two types of losses: technical losses (TLs), inherent to energy distribution and transformation, and non-technical losses (NTLs), which are related to energy fraud and metering errors. Although the supplied and recorded energy should ideally match, discrepancies due to systemic losses are inevitable \cite{frederiks2015household}.

\subsection{Technical Losses (TLs)}

TLs in electrical distribution systems result from energy dissipation in electrical components during transmission and distribution, contributing to increased costs and carbon emissions \cite{hertwich2019material}. Influencing factors include the physical properties of the electrical equipment, voltage levels, and grid design. Common sources of TLs encompass transformer winding losses and resistive losses in feeders and networks \cite{aminifar2021review}.

According to Congres International des Reseaux Electriques de Distribution \cite{CIREDDistributionLosses2017}, TLs primarily arise from load losses (variable resistive and reactance losses), no-load losses (fixed losses), and losses due to network services. Advancements in Information and Communication Technology (ICT) and data acquisition have enabled more precise computation and verification of TLs, as demonstrated by predictive models using transformer and SM data \cite{I2022transformer}.

\subsection{Non-Technical Losses (NTLs)}

NTLs, which occur independently of TLs, are often more challenging to measure and attribute to external factors such as energy theft and meter irregularities \cite{saeed2020detection}. These losses have significant implications for political stability and the financial well-being of UPs. High NTLs are commonly linked to governance issues like political instability and corruption \cite{de2021non}. Addressing NTLs is crucial for enhancing the efficiency of electrical distribution systems and reducing costs.

Key causes of NTLs include:

\begin{itemize}

\item Non-payment by consumers: Leading to systemic financial issues for UPs.

\item Meter irregularities: Resulting from incorrect meter readings and equipment malfunctions.

\item Energy theft: Entailing illegal connections and meter tampering.

\end{itemize}

\subsubsection{Non-Payment by Consumers}

Failure to pay utility bills can lead to a cascade of financial and systemic problems \cite{busby2021cascading}. Strategies like service disconnection and prepayment meters are implemented to mitigate these issues \cite{kambule2022formulating}.

\subsubsection{Meter Irregularities}

Meter irregularities, contributing significantly to NTLs, occur when meters inaccurately record energy consumption. Common causes include incorrect readings and equipment malfunctions (Tenaga Nasional Berhad, 2018). With AMI, UPs can effectively monitor SMs in real-time, aiding in the detection and resolution of these issues. This comprehensive monitoring enables UPs to identify discrepancies in energy reporting and manage the distribution system more efficiently.

Addressing NTLs is vital for the effective operation of electrical distribution systems, enhancing network efficiency, reducing costs, and improving reliability. Effective strategies involve ensuring accurate meter readings, proper installation and maintenance, and implementing systems to detect and prevent energy theft.

\subsection{Addressing Energy Theft}

Meter tampering and energy theft are primary contributors to Non-Technical Losses (NTLs) in Malaysia, as identified by Tenaga Nasional Berhad (2006). NTLs have persistently troubled Utility Providers (UPs) globally since the inception of energy billing. Current data suggests that worldwide energy theft amounts to an astonishing \\$96 billion annually (Northeast Group, 2017). In the United States, energy theft results in annual losses of about \$6 billion (Karaim, 2015), while UPs in India face yearly losses of around \$4.5 billion due to energy fraud (Ahmad et al., 2018). In Canada, British Columbia Hydro incurs yearly losses of \$100 million, largely attributed to illegal marijuana cultivation operations (Meuse,

2016). The increasing sophistication of energy theft techniques poses a significant challenge in detecting NTLs, leading to inflated costs for consumers and substantial government subsidies. This thesis, therefore, focuses on identifying and combating energy theft and meter irregularities in Smart Grids (SGs).

Various methods are employed to under-report energy usage:

\begin{itemize}

\item At the consumer level, common tactics include tampering with meters or siphoning energy from unoccupied premises.

\item At the grid level, fraudulent practices often involve bypassing meters by directly connecting high-load appliances or the entire electrical system to the feeder with an unauthorized distribution transformer (DT).

\item At the utility level, inaccuracies in billing, whether due to unintentional meter errors or intentional manipulation by corrupt personnel, can lead to profit losses.

\end{itemize}

To counteract energy theft, TNB Malaysia has formed specialized teams for physical meter inspections (Tenaga Nasional Berhad, 2006). These teams, equipped with additional technicians and resources, conduct investigations into suspected cases of energy fraud.

The implementation of Advanced Metering Infrastructure (AMI) and Smart Meters (SMs) is a strategic response to energy theft. SMs are engineered to detect and report tampering, mitigating certain vulnerabilities present in traditional analog meters. TNB's pilot project in 2015 involved installing 1,000 SMs in Malacca and Putrajaya (Tenaga Nasional Berhad, 2017), aiming to offer enhanced digital services, including failure alarms and tamper detection for analyzing NTLs.

However, the integration of smart metering infrastructure introduces new vulnerabilities. For instance, SMs are not immune to tampering (McLaughlin et al., 2010). An energy thief could gain unauthorized access to a SM, disrupting its communication and neutralizing automated alarms. Additionally, the high rate of false positives in meter alarms complicates the UPs' task of differentiating between fraudulent and legitimate consumers.

Literature review (McLaughlin et al., 2010, 2013; Jiang et al., 2014; Accenture, 2011; Y. Liu et al., 2018; Tellbach & Li, 2018) categorizes energy theft techniques into three main types:

\begin{enumerate}

- \item Physical attacks
- \item Cyberattacks
- \item Data attacks

\end{enumerate}

Data attacks can result from both cyber and physical breaches, compromising consumer usage data at various stages: recording, transmission, or storage (Xiao et al., 2013). These techniques are summarized in Section 2.4 below, which will be referenced in Section 4.3 to construct an attack model encompassing various known energy theft methods. Subsequently, energy theft and meter irregularity scenarios will be simulated by altering benign SM readings to assess the proposed anomaly detection frameworks in Chapters 5 \& 6.

\section{Classification of Attacks in Conventional and Smart Grid Systems}

\subsection{Physical Attacks}

These attacks involve direct interference with the electrical metering and distribution equipment.

- \begin{enumerate}
 - \item Meter Swapping: Replacement of meters with units from unoccupied or low-usage locations.
 - \item Power Diversion: Redirecting power supply within a neighborhood.
 - \item Meter Tampering: Includes meter removal/disconnection, reversing the meter's position, using magnets to affect readings, obstructing or damaging the rotating coil, introducing substances to impair meter function, unauthorized access to Smart Meters (SMs), and modifying the current transformer (CT) ratio.
 - \item Meter Bypass: Direct wiring of high-load appliances to the grid, bypassing the meter completely.
 - \item Illicit Grid Connection: Unauthorized connections to the primary voltage grid or distribution feeder.
 - \item Corruption: Bribery of utility personnel for altered billing.
 - \item Meter Calibration and Regulation Issues: Inappropriate calibration and regulation of meters.
- \end{enumerate}

\subsection{Cyber Attacks}

Cyber attacks target the digital and networked components of the grid.

\begin{enumerate}

- \item Credential Theft: Unauthorized access to meters using stolen login credentials.
 - \item Firmware Hacking: Remotely hacking into Smart Meter firmware.
 - \item Data Tampering: Altering meter storage data including total energy consumption, audit logs, and encryption keys.
 - \item Network Exploitation: Compromising meter readings through network vulnerabilities and intercepting meter communications.
 - \item Bandwidth Flooding: Overloading the Neighborhood Area Network (NAN) bandwidth.
 - \item Resource Exhaustion: Depleting meter memory or central processing unit (CPU) capacity.
 - \item Event Logging Interference: Deleting or interrupting logged events.
 - \item Communication Disruption: Disrupting radio frequency (RF) communications.
 - \item Value Injection: Injecting forged values into communications between Utility Providers (UPs) and Smart Meters.
 - \item Traffic Modification: Altering traffic between UPs and Smart Meters.
 - \item Meter Spoofing and Jamming: Impersonation of meters and RF signal jamming.
 - \item Malware: Designing and injecting malware into Smart Meters.
 - \item Pricing Manipulation: Altering predictive pricing algorithms.
- \end{enumerate}

\subsection{Data Attacks}

These attacks involve manipulation of consumption data.

- \begin{enumerate}
- \item Zero/Negative Reporting: Falsely reporting zero or negative energy consumption.
- \item Consumption Report Alteration: Stopping or altering energy usage reports.
- \item Measurement Exclusion: Removing high-consumption appliances from measurements.
- \item Under-reporting: Reporting less energy consumption than actual.
- \item Load Profile Modification: Changing appliance load profiles to hide larger loads.
- \end{enumerate}

\section{Strategies for Energy Theft}

From the outset of energy billing, fraudulent consumers have employed numerous methods to alter meters and their inputs for energy theft. These methods fall primarily into two categories: meter tampering, involving manipulation of the meter's internal structure, and line tampering, which includes bypassing the meter connection. This thesis focuses on identifying such attacks under the assumption that the integrity of meter consumption readings has been compromised, and does not address methods of manipulating communication signals.

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\begin{figure}[h!]  
  \centering  
  \includegraphics[width=1.0\textwidth]{src/Direct_tapping.pdf}  
  \caption{Direct tapping to the power line.}  
  \label{fig:theft}  
\end{figure}
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\textbf{Low Voltage Meters (230V Single Phase):}
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\begin{enumerate}
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  \item \textbf{Direct Connection to Power Grid (Bypassing the Meter):}
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Consumers, particularly in domestic or SME sectors, often directly tap into the LV 230V single-phase/415V three-phase power lines. This method is more straightforward and considered safer compared to tampering with HV lines.

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  \item \textbf{Meter Tampering:}
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  \begin{itemize}
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    \item Breaking the meter's enclosure seal for internal access and tampering.
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    \item Reversing number dials in analog meters or modifying CT turns.
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    \item Using magnets to slow down the rotor disk, affecting consumption measurement.
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    \item Installing remote control switches to regulate energy consumption recording.
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  \end{itemize}
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  \item \textbf{Circuit Bypass/Hidden Switch:}
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Utilizing jumper wires or hidden switches to bypass the metering circuit, avoiding energy consumption registration. This method is becoming increasingly sophisticated with the use of advanced gadgets and wiring.

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These strategies illustrate the evolving nature of energy theft, ranging from simple mechanical tampering to sophisticated electronic interference. The challenge for utility providers lies in detecting and mitigating these diverse methods of energy theft, ensuring accurate billing and fair energy distribution.

High Voltage (HV) meters, particularly three-phase watt-hour meters, are commonly installed at industrial sites with substantial energy demands. These meters typically employ a "two-watt-hour meters" connection in a three-phase energy meter for measurement. Energy theft in these settings, particularly prevalent in commercial and industrial sectors, contributes significantly to Non-Technical Losses (NTLs) in countries like Malaysia.

High Voltage Meters (12kV or 24kV three-phase, three or four-wire primary):

Enumerate

Direct Power Line Connection: Direct tapping into HV power lines is more challenging, with heightened risks due to the high voltage.

Meter Tampering: Similar methods as those employed in LV meters are applied, including physical alteration of the meter's components.

Control Wire Tampering: Energy theft executed by manipulating the control wires of the CT, resulting in reduced current readings.

Terminal/Meter Seal Tampering: Tampering with terminal seals located below the meter to alter voltage phase readings.

Voltage Tap Interference: Disrupting voltage readings by shorting voltage taps to the ground.

End Enumerate

NTL Detection Methods

2.4 NTL Detection Methods

Several recent studies have proposed various NTL detection methods, such as Support Vector Machine (SVM), load profiling, neural networks, and decision trees (Viegas et al., 2017). This thesis categorizes these detection schemes into three groups: state-based, game theory-based, and classification-based (Jiang et al., 2014), focusing primarily on data-oriented schemes that rely on consumer data like energy consumption measurements.

\subsection{State-based Detection:}

State-based detection involves monitoring the power system's state using specific equipment to identify energy fraud.

Xiao et al. (2013) proposed three inspection schemes for identifying anomalous Smart Meters (SMs), which, while effective, increase cost due to the additional metering equipment required.

McLaughlin et al. (2013) designed an Advanced Metering Infrastructure Intrusion Detection System (AMIDS) using various data sources, though this raises privacy concerns due to non-intrusive load monitoring (NILM).

Selvapriya (2014) suggested using control units to compare individual and aggregated consumption against feeder input levels, with alerts sent via GSM to investigate suspected energy theft.

Khoo and Cheng (2011) integrated RFID technology for better meter inventory management and energy theft reduction.

Huang et al. (2013) and Sahoo et al. (2015) used variance analysis and temperature-dependent predictive models, respectively, to identify abnormal energy consumption patterns.

Various recent studies have proposed NTL detection methods, focusing primarily on data-oriented schemes that rely on consumer data like energy consumption measurements.

Game-theoretic approaches to NTL detection conceptualize the interaction between fraudulent consumers and Utility Providers (UPs) as a strategic game (Cardenas et al., 2012; Amin et al., 2015). In this framework, energy thieves aim to under-report electricity consumption while minimizing detection risk. Conversely, UPs focus on enhancing the likelihood of identifying theft while also aiming to reduce the operational costs associated with these detection mechanisms. This perspective offers a novel way to analyze and mitigate NTLs.

In the game-theoretic framework proposed by Amin et al. (2015), two scenarios are considered: perfect competition and unregulated monopoly. The study develops an elaborate game-theoretic model to assess various statistical techniques for detecting energy theft. However, this model necessitates certain impractical assumptions about the nature of fraudulent activities. The research yields precise estimates of detection capabilities under these specific assumptions.

Cardenas et al. (2012) also developed a game-theoretic model, formulating a strategic interaction between energy thieves and UPs. They determined the Nash equilibrium of this game as the probability density function that both defenders and attackers should adopt when sending Advanced Metering Infrastructure (AMI) measurements. Furthermore, the study includes a preliminary analysis of the optimal sampling interval for Smart Meters (SMs), balancing consumer privacy concerns with the retention of demand response program benefits.

However, the development of utility functions for all involved parties, including energy thieves and UPs, and the formulation of their potential strategies remain complex challenges within this approach.

Game-theoretic approaches to NTL detection conceptualize the interaction between fraudulent consumers and Utility Providers (UPs) as a strategic game. In this framework, energy thieves aim to under-report electricity consumption while minimizing detection risk, whereas UPs focus on enhancing the likelihood of identifying theft and reducing operational costs.